

Revision Notes for Class 7 Maths Chapter 2 Arithmetic Expressions

In mathematics, simple expressions use numbers and basic operations like addition, subtraction, multiplication and division. These are known as arithmetic expressions. For example, " $13 + 2$ " or " 12×5 " are arithmetic expressions, and each such expression can be worked out to find its value. For instance, " $13 + 2$ " equals 15. We use the "=" sign to show that both sides have the same value.

Different Ways to Write Numbers

There are many ways to write the same number using arithmetic expressions. For example, the number 12 can be written as " $10 + 2$ ", " 3×4 ", or " $24 \div 2$ ". Using basic operators (+, -, $\times$, $\div$), one can express a number in several ways. Try this method with your favourite number and see how many expressions you can create. This builds fluency and confidence in handling numbers.

Comparing Expressions

Expressions can be compared by finding out which one is greater, less or equal. We use the symbols "=", "<", and ">" to compare values. For example, " $10 + 2 > 7 + 1$ " because 12 is greater than 8. In some questions, you can fill in missing numbers, such as " $13 + 4 = ____ + 6$ ", where the blank should be 11 to keep both sides equal ($13 + 4 = 17$; so $11 + 6 = 17$).

You can also arrange a set of expressions in increasing or decreasing order based on their evaluated values. For example, you may need to arrange " $67 - 19$ ", " 5×11 ", and " $120 \div 3$ " in order.

Smart Comparing Without Calculations

Sometimes, you do not need to actually work out each expression to compare them. For example, between " $1023 + 125$ " and " $1022 + 128$ ", look at how much each side changes: Joy started with one less than Raja, but got three more marbles, so Joy ends up with two more. This technique saves time in tests and improves logical thinking.

Evaluating Complex Expressions

Sometimes, expressions may have more than one operation, or lack context, leading to confusion about which operation to do first. For instance, " $30 + 5 \times 4$ " could be mistaken if a student adds first, but the convention is to perform multiplication before addition unless brackets are used to show otherwise. So, " $30 + 5 \times 4$ " should be calculated as $30 + (5 \times 4) = 30 + 20 = 50$.

Use of Brackets

Brackets help remove confusion and clarify what to do first. For example, " $100 - (15 + 56)$ " means that you add 15 and 56 first (getting 71), then subtract the result from 100. So, $100 - 71 = 29$. Without brackets, one might accidentally do the subtraction first which would give a wrong answer.

Terms in Expressions

Terms are parts of expressions separated by a plus sign (+). In the expression " $12 + 7$ ", the terms are 12 and 7. In " $83 - 14$ ", you can think of it as $83 + (-14)$, so the terms are 83 and -14. Knowing the terms helps when using properties like commutative and associative, or removing/process brackets.

Expression	As sum of terms	Terms
$13 - 2 + 6$	$13 + (-2) + 6$	13, -2, 6
$5 + 6 \times 3$	$5 + (6 \times 3)$	5, (6×3)
$4 + 15 - 9$	$4 + 15 + (-9)$	4, 15, -9
$23 - 2 \times 4 + 16$	$23 + (-2 \times 4) + 16$	23, -8, 16
$28 + 19 - 8$	$28 + 19 + (-8)$	28, 19, -8

Commutative and Associative Properties

Changing the order or grouping of terms while adding doesn't change the overall sum. This is called the commutative property (swapping) and associative property (grouping). For example, $(2 + 3) + 4$ is the same as $2 + (3 + 4)$.

Expressions Involving Multiplication and Division

When expressions have both addition and multiplication, the multiplication is done first unless brackets say otherwise. For example, in " $5 + 6 \times 3$ ", we first do $6 \times 3 = 18$, then add 5, so the answer is 23. However, in " $(5 + 6) \times 3$ ", we first add and then multiply, so the answer is 33. Brackets are important for guiding the order.

Real-life Examples

Mathematical expressions are useful in real life. For example, if Mallika spends ₹25 daily on lunch for 5 days, the total is " $5 \times 25 = 125$ ". For a game played with 33 students, grouping into 5 per group, the expression could be " $6 \times 5 + 3$ " (six groups of 5, plus 3 left). If you break 100 kg of rice into packets of 2 kg each, already having 4 packets, the total number is " $4 + (100 \div 2) = 54$ ".

Removing Brackets and Signs

To remove brackets that are preceded by a minus sign, change the operation signs inside the bracket to their opposites. For example, " $200 - (40 + 3)$ " equals " $200 - 40 - 3$ ". For " $500 - (250 - 100)$ ", apply the rule: it becomes " $500 - 250 + 100$ ". This skill helps solve questions with multi-step calculations with fewer mistakes.

Distributive Property

This property says that you can multiply a bracket by splitting it up: " $2 \times (43 + 24)$ " can be written as " $2 \times 43 + 2 \times 24$ ". Similarly, " $(4 + 3) \times 5$ " can be written as " $4 \times 5 + 3 \times 5$ ", and both give the same result. This property is often used in mental maths and shortcuts.

Expression Challenges and Activities

Try creating different expressions using given numbers and operations to reach certain values—for example, making as many numbers as possible using three 3's or four 4's.

Such challenges help sharpen calculation skills and deepen understanding of how arithmetic expressions are built and manipulated.

- Arithmetic expressions use numbers and operations ($+$, $-$, $\times$, $\div$) to create values.
- Symbols like "=", "<", ">" help compare expressions and their results.
- Brackets show which parts to calculate first.
- Terms are the separated parts in an expression, usually split by "+".
- Properties like commutative, associative, and distributive help to simplify and rearrange expressions without changing the result.
- Removing brackets requires attention to operation signs—to avoid errors, change the sign if a negative is outside the bracket.

Quick Revision Table

Property	Description	Example
Commutative (Addition)	Order does not matter	$8 + 6 = 6 + 8$
Associative (Addition)	Grouping does not matter	$(2 + 3) + 4 = 2 + (3 + 4)$
Distributive	Multiply outside by each term inside	$2 \times (7 + 5) = 2 \times 7 + 2 \times 5$

Practicing arithmetic expressions sharpens your calculation skills, logical thinking, and your ability to solve real-world problems efficiently. Remember to use brackets whenever needed, check your terms, and try different properties for quick and accurate results!